☐ High Priority

CONTRACTOR SUBMITTAL TRANSMITTAL FORM REV. A

Owner:	County of Hawaii		
Contractor:	Nan, Inc.	**Project No.:**	WW-4705R
Project Name:	Hilo WWTP Phase 1	**Submittal Number:**	03600-001.0
Submittal Title:	Product Data: Sika non-shrink Grout		☐ For Information Only
TO:	County of Hawaii, DEM		
From:	Nan Inc.		

Specification No. and Subject of Submittal / Equipment Supplier			
Spec: 03600	**Paragraph:** 1.03C(1), 1.03D(1), 2.01A, 2.01B.		
Authored By: Krishna Dubbudu		**Date Submitted:**	10/01/2025

Submittal Certification

Check Either (A) or (B):

☑ (A) We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings with no exceptions.

☐ (B) We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings except for the deviations listed.

Certification Statement: By this submittal, I hereby represent that I have determined and verified all field measurements, field construction criteria, materials, dimensions, catalog numbers and similar data, and I have checked and coordinated each item with other applicable approved shop drawings and all Con[illegible]

General Contractor's Reviewer's Signature: Krish

Printed Name and Title: Krishna Dubbudu, Project Engineer

In the event, Contractor believes the Submittal response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.

Firm: **Signature:** **Date Returned:**

PM/CM Office Use

Date Received GC to PM/CM:

Date Received PM/CM to Reviewer:

Date Received Reviewer to PM/CM:

Date Sent PM/CM to GC:

Nan, Inc

PROJECT: HILO WWTP REHABILITATION AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

THIS SUBMITTAL HAS BEEN CHECKED BY THIS CONTRACTOR. IT IS CERTIFIED CORRECT, COMPLETE, AND IN COMPLIANCE WITH CONTRACT DRAWINGS AND SPECIFICATIONS. ALL AFFECTED CONTRACTORS AND SUPPLIERS ARE AWARE OF, AND WILL INTEGRATE THIS SUBMITTAL (UPON APPROVAL) INTO THEIR OWN WORK.

DATE RECEIVED 10/2/2025
SPECIFICATION SECTION # 03600
SPECIFICATION Grouting
PARAGRAPH 1.03-C&D
DRAWING N/A
SUBCONTRACTOR N/A
SUPPLIER N/A
MANUFACTURER Sika Corp

CERTIFIED BY CQCM or Designee : Matthew Chun

SECTION 03600

GROUTING

PART 1 GENERAL

1.01 SUMMARY

A. Section includes:
   1. Cement grout.
   2. Cement mortar.
   3. Dry-pack mortar.
   4. Epoxy grout.
   5. Grout.
   6. Non-shrink epoxy grout.
   7. Non-shrink grout.

1.02 REFERENCES

A. ASTM International (ASTM):
   1. C109 - Standard Test Method for Compressive Strength of Hydraulic Cement Mortars (using 2-inch cube specimens).
   2. C230 - Standard Specification for Flow Table for Use in Tests of Hydraulic Cement.
   3. C531 - Standard Test Method for Liner Shrinkage and Coefficient of Thermal Expansion of Chemical-Resistant Mortars, Grouts, Monolithic Surfacings, and Polymer Concretes.
   4. C579 - Standard Test Method for Compressive Strength of Chemical-Resistant Mortars, Grouts, and Monolithic Surfacings and Polymer Concretes.
   5. C939 - Standard Test Method for Flow of Grout for Preplaced-Aggregate Concrete (Flow Cone Method).
   6. C942 - Standard Test Method for Compressive Strength of Grouts for Preplaced-Aggregate Concrete in the Laboratory.
   7. C1107 - Standard Specification for Packaged Dry, Hydraulic-Cement Grout (Non-shrink).
   8. C1181 - Standard Test Methods for Compressive Creep of Chemical-Resistant Polymer Machinery Grouts.

B. International Concrete Repair Institute (ICRI):
   1. 310.2R - Selecting and specifying Concrete Surface Preparations for Sealers, Coatings, Polymer Overlays, and Concrete Repair.

1.03 SUBMITTALS

A. Cement grout: Not Included
   1. Mix design.
   2. Material submittals.

B. Cement mortar: Not Included
1. Mix design.

C. Material submittals. Non-shrink epoxy grout: Page 10-16
1. Manufacturer's literature.
2. Quality control tests results. Can only be provided after the approval

D. Non-shrink grout:
1. Manufacturer's literature. Page 17-21
2. Quality control tests results. Can only be provided after the approval

1.04 DELIVERY, STORAGE, AND HANDLING

A. Deliver materials to jobsite in their original, unopened packages or containers, clearly labeled with manufacturer's product identification and printed instructions.

B. Store materials in cool dry place and in accordance with manufacturer's recommendations.

C. Handle materials in accordance with the manufacturer's instructions.

PART 2 PRODUCTS

2.01 MANUFACTURED UNITS

A. Non-shrink epoxy grout:
1. Manufacturers: One of the following or equal:
   a. Five Star Products, Inc., Five Star DP Epoxy Grout.
   b. Master Builder Solutions, MasterFlow 648. Note: Product Changed name to Sika 648
   c. L&M Construction Chemicals, Inc., EPOGROUT.
2. Non-shrink epoxy grout shall be 100 percent solid, premeasured, prepackaged system containing 2-component thermosetting epoxy resin and inert aggregate.
3. Maintain flowable consistency for at least 45 minutes at 70 degrees Fahrenheit.
4. Shrinkage or expansion: Less than 0.0006 inches per inch when tested in accordance with ASTM C531.
5. Minimum compressive strength: 10,000 pounds per square inch at 24 hours and 14,000 pounds per square inch at 7 days when tested in accordance with ASTM C579, Method B.
6. Compressive creep: Not exceed 0.0037 inches/per inch when tested under 400 pounds per square inch constant load at 140 degrees Fahrenheit in accordance with ASTM C1181.
7. Coefficient of thermal expansion: Not exceed 0.000018 inches per inch per degree Fahrenheit when tested in accordance with ASTM C531, Method B.

Note: Product Changed name to Sika 928

B. Non-shrink grout:
1. Manufacturers: One of the following or equal:
   a. Five Star Products, Inc., Five Star Grout.
   b. Master Builder Solutions, MasterFlow 928.
   c. L&M Construction Chemicals, Inc., CRYSTEX.
2. In accordance with ASTM C1107.
3. Preportioned and prepackaged cement-based mixture.
4. Contain no metallic particles such as aluminum powder and no metallic aggregate such as iron filings.
5. Require only addition of potable water.
6. Water for pre-soaking, mixing, and curing: Potable water.
7. Free from emergence of mixing water from within or presence of water on its surface.
8. Remain at minimum flowable consistency for at least 45 minutes after mixing at 45 degrees Fahrenheit to 90 degrees Fahrenheit when tested in accordance with ASTM C230.
   a. If at fluid consistency, verify consistency in accordance with ASTM C939.
9. Dimensional stability (height change):
   a. In accordance with ASTM C1107, volume-adjusting Grade B or C at 45 degrees Fahrenheit to 90 degrees Fahrenheit.
   b. Have 90 percent or greater bearing area under bases.
10. Have minimum compressive strengths at 45 degrees Fahrenheit to 90 degrees Fahrenheit in accordance with ASTM C1107 for various periods from time of placement, including 5,000 pounds per square inch at 28 days when tested in accordance with ASTM C109 as modified by ASTM C1107.

2.02 MIXES

A. Cement grout:
1. Use same sand-to-cementitious materials ratio for cement grout mix that is used for concrete mix.
2. Use same materials for cement grout that are used for concrete.
3. Use water-to-cementitious materials ratio that is no more than that specified for concrete.
4. For spreading over surfaces of construction or cold joints.

B. Cement mortar:
1. Use same sand-to-cementitious materials ratio for cement mortar mix that is used for concrete mix.
2. Use same materials for cement mortar that are used for concrete.
3. Use water-to-cementitious materials ratio that is no more than that specified for concrete being repaired.
4. At exposed concrete surfaces not to be painted or submerged in water: Use sufficient white cement to make color of finished patch match that of surrounding concrete.

C. Dry-pack mortar:
1. Proportions by weight: 1 part portland cement to 2 parts concrete sand.
   a. Portland cement: As specified in Section 03300 - Cast-in-Place Concrete.
   b. Concrete sand: As specified in Section 03300 - Cast-in-Place Concrete.

D. Epoxy grout:
   1. Consist of mixture of epoxy or epoxy gel and sand.
      a. Epoxy: As specified in Section 03071 - Epoxies.
      b. Epoxy gel: As specified in Section 03071 - Epoxies.
      c. Sand: Clean, bagged, graded, and kiln-dried silica sand.
   2. Proportioning:
      a. For horizontal work: Consist of mixture of 1 part epoxy with not more than 2 parts sand.
      b. For vertical or overhead work: Consist of 1 part epoxy gel with not more than 2 parts sand.

E. Grout:
   1. Mix in proportions by weight: 1 part portland cement to 4 parts concrete sand.
      a. Portland cement: As specified in Section 03300 - Cast-in-Place Concrete.
      b. Concrete sand: As specified in Section 03300 - Cast-in-Place Concrete.

F. Non-shrink epoxy grout:
   1. Mix in accordance with manufacturer's installation instructions.

G. Non-shrink grout:
   1. Mix in accordance with manufacturer's installation instructions such that resulting mix has flowable consistency and is suitable for placing by pouring.

PART 3 EXECUTION

3.01 EXAMINATION

A. Inspect concrete surfaces to receive grout or mortar and verify that they are free of ice, frost, dirt, grease, oil, curing compounds, paints, impregnations, and loose material or foreign matter likely to reduce bond or performance of grout or mortar.

3.02 PREPARATION

A. Surface preparation for grouting other baseplates:
   1. Remove grease, oil, dirt, dust, curing compounds, laitance, and other deleterious materials that may affect bond to concrete and bottoms of baseplates.
   2. Roughen concrete surfaces in contact with grout to ICRI CSP-6 surface profile or rougher.
      a. Remove loose or broken concrete.
   3. Metal surfaces in contact with grout: Grit blast to white metal surface.

3.03 INSTALLATION

A. Mixing:
   1. Cement grout:
      a. Use mortar mixer with moving paddles.
      b. Pre-wet mixer and empty out excess water before beginning mixing.

2. Cement mortar:
   a. Use mortar mixer with moving paddles.
   b. Pre-wet mixer and empty out excess water before beginning mixing.
3. Dry-patch mortar:
   a. Use only enough water so that resulting mortar will crumble to touch after being formed into ball by hand.
4. Non-shrink epoxy grout:
   a. Keep temperature of non-shrink epoxy grout from exceeding manufacturer's recommendations.
5. Non-shrink grout:
   a. May be drypacked, flowed, or pumped into place. Do not overwork grout.
   b. Do not retemper by adding more water after grout stiffens.

B. Placement:
1. Cement grout:
   a. Exercise care in placing cement grout because it is required to furnish structural strength, impermeable water seal, or both.
   b. Do not use cement grout that has not been placed within 30 minutes after mixing.
2. Cement mortar:
   a. Use mortar mixer with moving paddles.
   b. Pre-wet mixer and empty out excess water before beginning mixing.
3. Epoxy grouts:
   a. Wet surfaces with epoxy for horizontal work or epoxy gel for vertical or overhead work prior to placing epoxy grout.
4. Non-shrink epoxy grout:
   a. Mix in complete units. Do not vary ratio of components or add solvent to change consistency of mix.
   b. Pour hardener into resin and mix for at least 1 minute and until mixture is uniform in color. Pour epoxy into mortar mixer wheelbarrow and add aggregate. Mix until aggregate is uniformly wetted. Over mixing will cause air entrapment in mix.
5. Non-shrink grout:
   a. Add non-shrink cement grout to premeasured amount of water that does not exceed the manufacturer's maximum recommended water content.
   b. Mix in accordance with manufacturer's instructions to uniform consistency.

C. Curing:
1. Cement based grouts and mortars:
   a. Keep continuously wet for minimum of 7 days. Use wet burlap, soaker hose, sun shading, ponding, and in extreme conditions, combination of methods.
   b. Maintain above 40 degrees Fahrenheit until it has attained compressive strength of 3,000 pounds per square inch, or above 70 degrees Fahrenheit for minimum of 24 hours to avoid damage from subsequent freezing.
2. Epoxy based grouts:
   a. Cure grouts in accordance with manufacturers' recommendations.
      1) Do not water cure epoxy grouts.
   b. Do not allow any surface in contact with epoxy grout to fall below 50 degrees Fahrenheit for minimum of 48 hours after placement.

D. Grouting equipment bases, baseplates, soleplates, and skids: As specified in Section 15050 - Common Work Results for Mechanical Equipment.

E. Grouting other baseplates:
   1. General:
      a. Use non-shrink grout as specified in this Section.
      b. Baseplate grouting shall take place from one side of baseplate to other in continuous flow of grout to avoid trapping air in grout.
      c. Maintain hydrostatic head pressure by keeping level of grout in headbox above bottom of baseplate. Fill headbox to maximum level and work grout down.
      d. Vibrate, rod, or chain non-shrink grout to facilitate grout flow, consolidate grout, and remove trapped air.
   2. Forms and headboxes:
      a. Build forms using material with adequate strength to withstand placement of grouts.
      b. Use forms that are rigid and liquidtight. Caulk cracks and joints with elastomeric sealant.
      c. Line forms with polyethylene for easy grout release. Coating forms with 2 coats of heavy-duty paste wax is also acceptable.
      d. Headbox shall be 4 to 6 inches higher than baseplate and shall be located on one side of baseplate.
      e. After grout sets, remove forms and trim back grout at 45 degree angle from bottom edges of baseplate.

3.04 FIELD QUALITY CONTROL BY CONTRACTOR

A. Provide Contractor quality control as specified in Section 01450 - Quality Control.

B. Field inspections and tests:
   1. Non-shrink epoxy grout:
      a. Test for 24-hour compressive strength in accordance with ASTM C579, Method B.
      b. Three samples each placement.
   2. Non-shrink grout:
      a. Test for 24-hour compressive strength in accordance with ASTM C942.
      b. Three samples each placement.
   3. Submit records of inspections and test to Engineer within 24 hours after completion.

3.05 FIELD QUALITY CONTROL BY OWNER

A. Provide Owner quality control as specified in Section 01450 - Quality Control.

B. Special inspections, special tests, and structural observation:
   1. Not required.

C. Field inspections:
   1. Preparation.
      a. Review manufacturer's product data and installation instructions.

2. Required inspections:
   a. Observe surfaces to be injected for temperature and moisture conditions.
   b. Observe conditioning and preparation of urethane resin.
   c. Observe injection procedures for filling cracks.
3. Records of inspections:
   a. Provide record of each inspection.
   b. Submit to Engineer upon request.

END OF SECTION

PRODUCT DATA

BUILDING TRUST

PRODUCT DATA SHEET

SikaFlow®-648

(formerly MFlow 648)

High-strength, high-flow, chemical resistant epoxy grout

PRODUCT DESCRIPTION

SikaFlow®-648 is a three-component epoxy resin-based precision grout used to secure critical equipment for proper alignment and transmission of static and dynamic loads. With carefully balanced physical properties and excellent resistance to chemical attack, elevated service temperatures, vibration and torque, SikaFlow®-648 is formulated for easy installation, with good flow characteristics suitable for pouring or pumping in thicknesses up to 6", low dust generation and soap and water clean-up.

SikaFlow®-648 is available in all regions of the world, supported by trained sales and technical personnel with experience in the specification and installation of epoxy grouts on every continent.

USES

SikaFlow®-648 is used for assembling and fixing of the following items:

- Precision alignment of compressors, generators, pumps, fans and electric motors
- Pour-back grouting for post-tensioning cables
- Sole plates
- Crane rail grouting
- Grouting of rolling, stamping, grinding, crushing, drawing and finishing mills, forging hammers and other equipment subject to high torque, impact and vibration
- Grouting of wind turbine tower bases
- Grouting of anchors, bars and dowels

Note: For off-shore wind turbine installations please refer to our Sikagrout-9000 series.

SikaFlow®-648 is typically used in the following industries:

- Chemical processing
- Oil and Gas extraction, refining, processing and distribution
- Power generation
- LNG production, storage and transmission
- Pulp and paper production
- Steel and aluminum manufacturing
- Mining
- Other heavy industry

CHARACTERISTICS / ADVANTAGES

- High early and ultimate strengths for rapid turnaround
- Low creep maintains equipment alignment
- Retains physical properties at elevated temperatures
- Low-dusting for added worker comfort and safety
- Very low shrinkage for full baseplate contact
- Excellent flowability with high bearing area for even load distribution
- Variable fill ratio for desired flowability
- Excellent adhesion to steel and concrete for optimum load transfer and vibration dampening
- High chemical resistance
- Excellent freeze/thaw resistance for equipment in low temperature service environments
- Resists water and chloride intrusion for use in wet and aggressive environments
- Resists impact and dampens torque to protect equipment and extend service life
- Extended working time
- Pumpable for maximum productivity if needed
- Durable bond to concrete and steel optimizes load transfer
- Globally available for consistent project results.

PRODUCT INFORMATION

Packaging

54.4 lb unit (0.4 ft^3 [0.01 m^3])*

Part A	Resin	5.5 lb (2.5 kg) pail
Part B	Hardener	1.65 lb (0.75 kg) bottle
Part C	Aggregate	47 lb (21.3 kg) bag

*All components are packaged in a 6-gallon pail

216.6 lb unit (98.2 kg) (1.7 ft^3 [0.05 m^3])**

Part A	Resin	22.0 lb (10.0 kg) pail
Part B	Hardener	6.6 lb (3.0 kg) bottle
Part C	Aggregate	4 x 47 lb (21.3 kg) bags

**All components packaged separately. May be ordered as a 3-bag unit, which will yield 1.4 ft^3 (0.04 m^3). When estimating project requirements, be sure to account for application variables

Appearance / Color

Dark Grey

Shelf Life

24 months if stored at below mentioned storage conditions.

Storage Conditions

Store at ambient temperatures (60 to 80°F [16 to 27° C]), out of direct sunlight, in cool, dry conditions and clear of the ground on pallets protected from rainfall prior to application. The resin parts need to be protected from frost.

Density

4-Bag Mix (6.55:1)	130 lb/ft^3 (2,084 kg/m^3)	ASTM C 905
3-Bag Mix (4.92:1)	123 lb/ft^3 (1,971 kg/m^3)	

Flash Point

Part A (Resin)	419 °F (215 °C)	Pensky-Martens Closed Cup
Part B (Hardener)	325 °F (163 °C)	

TECHNICAL INFORMATION

Abrasion Resistance

Better than concrete

Impact Strength

Better than concrete

Compressive Strength

Consistency (Fill Ratio)	7 Day Ambient	Post Cured	
4-Bag Mix (6.55:1)	14,500 psi (100 MPa)	16,000 psi (110 MPa)	ASTM C 579, Method B
3-Bag Mix (4.92:1)	14,000 psi (96 MPa)	15,000 psi (103 MPa)	

Time	50 °F	77 °F	90 °F	
24 Hours	4,700 psi (32 MPa)	10,800 psi (75 MPa)	12,700 psi (88 MPa)	ASTM C 579, Method B
2 Days	10,000 psi (69 MPa)	12,100 psi (83 MPa)	13,200 psi (91 MPa)	
3 Days	11,000 psi (76 MPa)	13,000 psi (90 MPa)	13,600 psi (94 MPa)	
4 Days	12,000 psi (83 MPa)	13,400 psi (92 MPa)	13,900 psi (96 MPa)	

*Filled 6.55:1, cured 24 hours at room temperature, post cured 16 hours at 140 ºF, and conditioned 24 hours at test temperature.

Effective Bearing Area

1" Clearance	High (>85%)*	ASTM C 1339
2" Clearance	High (>85%)*	

*4-Bag Mix (6.55:1)

Flexural Strength

4-Bag Mix (6.55:1)	4,700 psi (32 MPa)	ASTM C 580, 73 °F (23 °C)
3-Bag Mix (4.92:1)	4,300 psi (30 MPa)	

Modulus of Elasticity in Flexure

4-Bag Mix (6.55:1)	2.5 x 10^6 psi (17.2 GPa)	ASTM C 580, 73 °F (23 °C)
3-Bag Mix (4.92:1)	2.3 x 10^6 psi (15.9 GPa)	

Tensile Strength

4-Bag Mix (6.55:1)	2,000 psi (13.8 MPa)	ASTM C 307
3-Bag Mix (4.92:1)	1,800 psi (12.4 MPa)	

Shear Strength

2,000 psi (14 MPa)* — Michigan DOT

*Bond strength to steel at 73 °F (23 °C)

Shrinkage

4-Bag Mix (6.55:1)	0.014	ASTM C 531
3-Bag Mix (4.92:1)	0.031	

unrestrained; linear, %,

Creep

Consistency (Fill Ratio)	400 psi (2.8 MPa) load	600 psi (4.1 MPa) load	
4-Bag Mix (6:55:1)	3.8 x 10^{-3}	4.2 x 10^{-3}	ASTM C 1181, 140° F (60° C), 28 days
3-Bag Mix (4:92:1)	4.0 x 10^{-3}	4.9 x 10^{-3}	

Coefficient of Thermal Expansion

4-Bag Mix (6.55:1)	18 x 10^{-6} in/in/°F (32.2 x 10^6 cm/cm/°C)	ASTM C 531, 73–210° F (23–99° C)
3-Bag Mix (4.92:1)	20 x 10^{-6} in/in/°F (36 x 10^6 cm/cm/°C)	

Flow Rate

Clearance	Back of Box	Full Plate Contact	
1"	11 min*	13 min*	ASTM C 580
2"	4 min, 45 sec*	5 min*	

*4-Bag Mix (6.55:1)

APPLICATION INFORMATION

Pot Life	Temperature	Time
	90 °F (32 °C)	50-60 min
	70 °F (21 °C)	90-120 min
	50 °F (10 °C)	120-150 min

BASIS OF PRODUCT DATA

Results may differ based upon statistical variations depending upon mixing methods and equipment, temperature, application methods, test methods, actual site conditions and curing conditions.

AVAILABILITY/WARRANTY

HANDLING AND TRANSPORT Usual preventive measures for the handling of chemical products should be observed when using this product, for example do not eat, smoke or drink while working and wash hands when taking a break or when the job is completed. Specific safety information referring the handling and transport of this product can be found in the Material Safety Data Sheet. For full information on Health and Safety matters regarding this product the relevant Health and Safety Data Sheet should be consulted.
Disposal of product and its container should be carried out according to the local legislation in force. Responsibility for this lies with the final owner of the product.

ENVIRONMENTAL, HEALTH AND SAFETY

This product is an article as defined in article 3 of regulation (EC) No 1907/2006 (REACH). It contains no substances which are intended to be released from the article under normal or reasonably foreseeable conditions of use. A safety data sheet following article 31 of the same regulation is not needed to bring the product to the market, to transport or to use it. For safe use follow the instructions given in the product data sheet. Based on our current knowledge, this product does not contain SVHC (substances of very high concern) as listed in Annex XIV of the REACH regulation or on the candidate list published by the European Chemicals Agency in concentrations above 0,1 % (w/w).

APPLICATION INSTRUCTIONS

Figure 1 - Regular Equipment

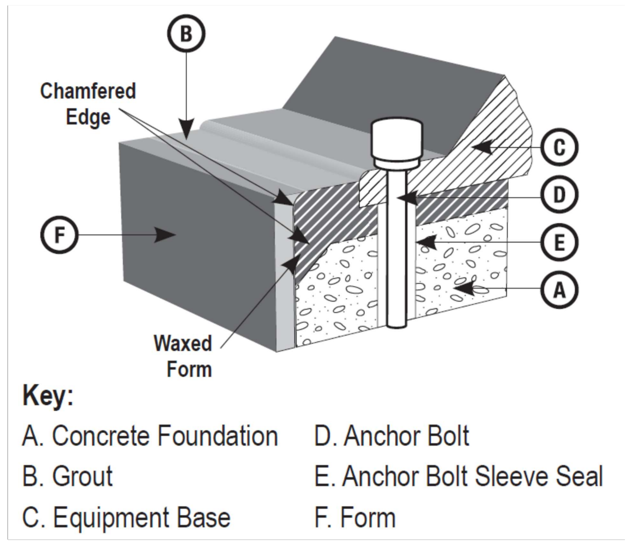


Figure 2 - Engine with Oil Pan

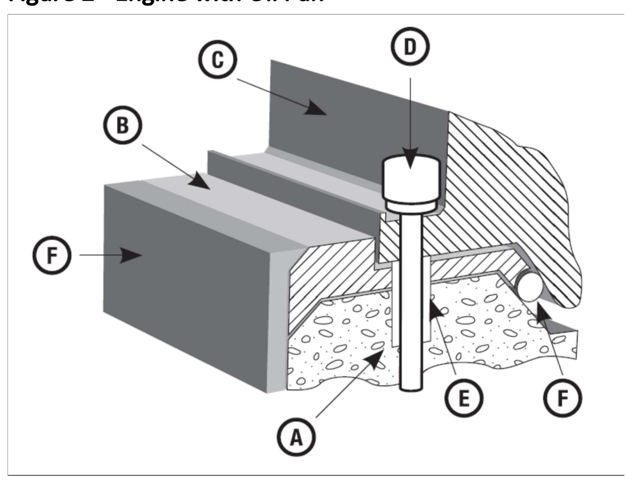


Figure 3 - Rail or Soleplate

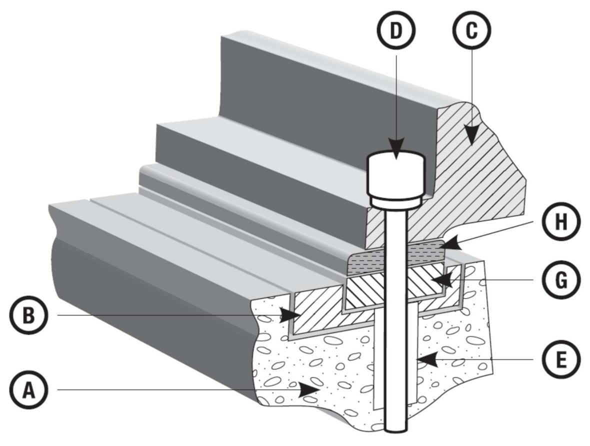


Figure 4 - Typical Rail with Expansion Joint Section

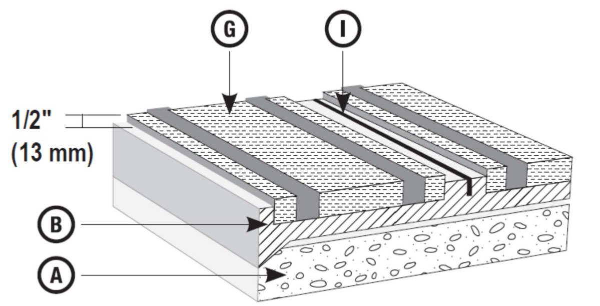
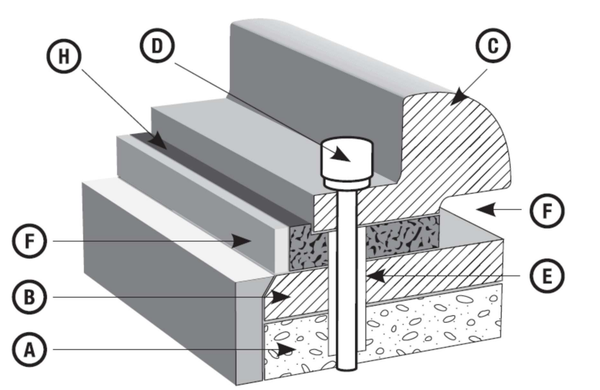


Figure 5 - Typical Epoxy Chock Application



NOTES ON INSTALLATION

- Do not add solvent, water, or any other material to the grout. Do not alter the resin or hardener proportions.
- Contact your local representative for a pre-job conference to plan the installation.
- For guidelines on specific anchor-bolt applications, contact Technical Service.
- Always use a head box when placing less than 1" (25 mm) depths.
- Substrate temperature must be greater than 50° F (10° C).
- Cold material will exhibit decreased flowability and reduced strength development.
- When using SikaFlow®-648 in post-tensioning anchorages, always use the standard aggregate load of 6.55:1.
- Minimum placement thickness is ½" (13 mm). Contact Technical Service before placing lifts more than 6" (152 mm) in depth.
- Chamfering the concrete edge helps reduce thermal cracking. Following proper installation procedures also reduce the potential for cracking.
- For professional use only; not for sale to or use by the general public.
- Make certain the most current versions of product data sheet and SDS are being used; visit https://usa.sika.com/ to verify the most current versions.
- Proper application is the responsibility of the user. Field visits by Sika personnel are for the purpose of making technical recommendations only and not for supervising or providing quality control on the jobsite.

EQUIPMENT

1. If rust scale is present, abrade the bonding surfaces of the base to be grouted; it must be free of coatings, wax, grease, or scale. Mechanical methods, such as grinding or sanding, will suffice, but do not produce as high a bond strength as sandblasting.
2. Primer should be used ONLY when a long delay between cleaning and grouting could allow excessive rusting or contamination. If the base must be primed, use Sikadur-32 Hi-Mod.
3. The grout should come up at least ¾" (19 mm) onto the equipment. Protect the area above it with masking tape.
4. To facilitate cleanup, wax or cover all surfaces where the grout may splash or spill.

SURFACE PREPARATION

1. Cure the foundation until design strength of the concrete is achieved and foundation is dry. Use the recommended procedure according to ACI 351.1R, Grouting Between Foundations and Bases for Support of Equipment and Machinery.
2. The surface to be grouted must be clean, strong, and roughened to a CSP of 5–9, following ICRI Technical Guideline No. 310.2 to permit proper bond. Do not use a bushing hammer.
3. Chamfer the edge of the concrete 45 degrees to about a 2" (51mm) width.
4. If an anchor bolt sleeve is to be filled, be sure all water is removed. Use a siphon, vacuum pump, or rubber hose and bulb. Remove the residual moisture by either forced air or evaporation.
5. Seal the anchor bolt hole with felt, foam rubber, or other means.
6. Cover all shims and leveling screws with putty or clay to keep the grout from adhering. Use model clay, glazing putty, or anything with a putty consistency that will stick but not harden. Shims or jack pockets may be

formed with wood, and forms filled with damp sand.
7. Remove shims or jack screws after the grout cures.
8. Shade the foundation from direct sunlight for at least 24 hours before and 48 hours after grouting.

MIXING

1. Aggregate must be completely dry.
2. Precondition all components to 70° F (21° C) for 24 hours before using.
3. Pour the hardener (Part B) into a pail of grout resin (Part A) and stir by hand with a spatula or paint stir paddle until well mixed to a uniform amber color.
4. Pour the mixture into a horizontal shaft mortar mixer or a Kol type mixer without delay.
5. Add the grout aggregate, one bag at a time, and mix only until aggregate is completely wetted out to avoid air entrapment. The first batch may be slightly less fluid than later batches because some of the resin is retained on the walls of the mixer. Withholding ½–1 bag of aggregate from the first batch of a full unit will compensate for lost resin. Note: always add aggregate to the mixer after the premixed liquids have been poured in.

Temperature	1.7 ft^3 unit very thin pours or very long distance	Standard pours
> 90° F (> 32° C)	—	—
70 to 90° F (21 to 32° C)	Up to ½ bag	—
50 to 70° F (10 to 21° C)	½ to 1 bag	½ bag

APPLICATION

Lengths of metal strapping laid in the formwork prior to placing may be necessary to assist grout flow over large areas and in compacting and eliminating air pockets. Have sufficient manpower, materials and tools to make mixing and placing rapid and continuous. Where grout must flow some distance, make the initial batch slightly more fluid or flowable than required; this lubricates the surfaces and avoids blockage of the grout that follows. The grout shall be poured continuously and from one side only, to avoid entrapment of air while grouting. Maintain a constant hydrostatic head, preferably of at least 4”. On the side where the grout has been poured, allow 4” clearance between the side of the form and the base plate of the machine. On the opposite side allow 2”- 4” clearance between the formwork and the base plate. Due to differences in temperature between the grout under the base plate, and exposed shoulders that are subject to more rapid temperature changes, debonding and / or cracking can occur. Avoid shoulders wherever possible. If shoulders are required, they should be firmly anchored with reinforcing to the substrate to prevent debonding.
Make sure grout fills the entire space to be grouted and remains in contact with the plate throughout the entire grouting placement.

Note: Do not use vibrator for placing the grout!

COLD-WEATHER CURING

For cold weather grouting use SikaFlow 640 Accelerator. Refer to the SikaFlow 640 Accelerator PDS.

1. The foundation and the equipment base will probably be cooler than room temperature unless room temperature has been constant for some time. Use the foundation and engine temperature, therefore, in estimating cure time.
2. Temperatures vary so radically, day vs. night, atmospheric vs. metal surface, that field judgment must still be used as the final measure. Cured grout should have a solid, almost metallic feel when struck with a hammer. Be sure to check as close to the base of the equipment as possible.

HOT-WEATHER GROUTING

1. Special care must be exercised when grouting at elevated temperatures, to reduce risks of premature hardening and subsequent cracking.
2. If the packaged grout is above 90° F (32° C), chill the sealed pails of grout resin in a tub of ice or cover the pails with water-soaked burlap to cool the grout to 70° F (21° C).
3. Provide shade from direct sunlight for at least 24 hours before and 48 hours after grouting.

COLD-WEATHER GROUTING

1. Temperatures below 60° F (16° C) make the grout stiff and hard to handle and significantly increase the cure time. The baseplate and foundation may be much cooler than room temperature. In cold weather, store materials in a warm place. For best handling, the temperature of the grout components and mixing equipment should be at least 70° F (21° C).
2. When baseplate and foundation temperatures (measured by a contact thermometer) are less than 50° F (10° C), heating of the area may be necessary.
3. If heating is required, erect an enclosure around the equipment and foundation to be grouted. Forced air or infrared heaters may be used to obtain the necessary heat to increase the baseplate and foundation temperatures to 50 to 70° F (10 to 21° C). Avoid local hot spots. Apply heat 1–2 days in advance of grouting to achieve uniform baseplate and foundation temperatures. Avoid exposure to exhaust from heating equipment. Remove heat during grout placement.
4. For temperatures from 40 to 50° F (4 to 10° C), consider using SikaFlow 640 Grout Accelerator to accelerate strength development.

CLEANING OF TOOLS

After the pour is complete, remove uncured epoxy from the mixer, wheelbarrow and tools with soap and water or a citrus degreaser. Cured material can only be removed mechanically.

OTHER RESTRICTIONS

See Legal Disclaimer.

LEGAL DISCLAIMER

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• NOT FOR INTERNAL CONSUMPTION
• FOR INDUSTRIAL USE ONLY
• FOR PROFESSIONAL USE ONLY

Prior to each use of any product of Sika Corporation, its subsidiaries or affiliates ("SIKA"), the user must always read and follow the warnings and instructions on the product's most current product label, Product Data Sheet and Safety Data Sheet which are available at usa.sika.com or by calling SIKA's Technical Service Department at 1-800-933-7452. Nothing contained in any SIKA literature or materials relieves the user of the obligation to read and follow the warnings and instructions for each SIKA product as set forth in the current product label, Product Data Sheet and Safety Data Sheet prior to use of the SIKA product.

SIKA warrants this product for one year from date of installation to be free from manufacturing defects and to meet the technical properties on the current Product Data Sheet if used as directed within the product's shelf life. User determines suitability of product for intended use and assumes all risks. User's and/or buyer's sole remedy shall be limited to the purchase price or replacement of this product exclusive of any labor costs. **NO OTHER WARRANTIES EXPRESS OR IMPLIED SHALL APPLY INCLUDING ANY WARRANTY OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE. SIKA SHALL NOT BE LIABLE UNDER ANY LEGAL THEORY FOR SPECIAL OR CONSEQUENTIAL DAMAGES. SIKA SHALL NOT BE RESPONSIBLE FOR THE USE OF THIS PRODUCT IN A MANNER TO INFRINGE ON ANY PATENT OR ANY OTHER INTELLECTUAL PROPERTY RIGHTS HELD BY OTHERS.**

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PRODUCT DATA

BUILDING TRUST

PRODUCT DATA SHEET

SikaGrout®-928

(formerly MFlow 928)

LOW DUST, HIGH-PRECISION MINERAL-AGGREGATE GROUT WITH EXTENDED WORKING TIME

PRODUCT DESCRIPTION

SikaGrout®-928 grout is a hydraulic cement-based mineral aggregate non-shrink grout with extended working time. It is ideally suited for grouting machines or plates requiring precision load-bearing support. It can be placed from fluid to damp pack over a temperature range of 45 to 90 °F (7 to 32 °C).

USES

- Grouting of equipment, such as compressors and generators, pump bases and drive motors, tank bases, and conveyors.
- Grouting anchor bolts, rebar, and dowel rods
- Grouting of precast wall panels, beams, columns, curtain walls, concrete systems, and other structural and non-structural building components
- Repairing concrete, including grouting voids and rock pockets

Substrates
- Concrete

CHARACTERISTICS / ADVANTAGES

- Meets the requirements of ASTM C1107 and US Army Corps of Engineers CRD C621 (Grades B and C) at a fluid consistency over a 30-minute working time
- Low-dusting for added worker comfort and safety
- NSF/ANSI 61 Std for use with potable water
- Pumpable
- Extended working time
- Can be mixed at a wide range of consistencies
- Freeze/thaw resistant making it suitable for exterior applications
- Hardens free of bleeding, segregation, or settlement shrinkage to provide maximum effective bearing area for optimum load transfer
- Contains high-quality, well-graded quartz aggregate for optimum strength and workability
- Sulfate resistant for marine, wastewater, and other sulfate-containing environments

APPROVALS / STANDARDS

- ASTM C 1107 and CRD 621, Grades B and C, requirements at a fluid consistency over a temperature range of 40–90 °F (4–32 °C)
- NSF/ANSI 61 Std for use with potable water

PRODUCT INFORMATION

Chemical Base	SikaGrout®-928 is a hydraulic cement-based mineral-aggregate grout.
Packaging	55 lb (25 kg) polyethylene-lined bags 3,300 lb (1,500 kg) bulk bags
Shelf Life	55 LB BAG: 1 year when properly stored 3,300 LB BULK BAG: 3 months when properly stored
Storage Conditions	Store in unopened containers in cool, clean, dry conditions

TECHNICAL INFORMATION

Specific Advice

Dust Reduction

SikaGrout®-928 vs Control	50%	(DIN55992-2)

Testing

Jobsite Testing

If strength tests must be made at the jobsite, use 2” (51 mm) metal cube molds as specified by ASTM C 942 and the ASTM C 1107 modification of ASTM C 109. DO NOT use cylinder molds. Control field and laboratory tests on the basis of desired placement consistency rather than strictly on water content.

Compressive Strength

	Plastic[1]	Consistency Flowable[2]	Fluid[3]
1 day	4,500 psi (31 MPa)	4,000 psi (28 MPa)	3,500 psi (24 MPa)
3 days	6,000 psi (41 MPa)	5,000 psi (34 MPa)	4,500 psi (31 MPa)
7 days	7,500 psi (52 MPa)	6,700 psi (46 MPa)	6,500 psi (45 MPa)
28 days	9,000 psi (62 MPa)	8,000 psi (55 MPa)	7,500 psi (52 MPa)

(ASTM C 942) according to ASTM C 1107 of ASTM C 109

1. 100–125% flow on flow table per ASTM C 230
2. 125–145% flow on flow table per ASTM C 230
3. 25 to 30 seconds through flow cone per ASTM C 939

Modulus of Elasticity in Compression

3 days	2.82×10^6 psi (1.94×10^4 MPa)
7 days	3.02×10^6 psi (2.08×10^4 MPa)
28 days	3.24×10^6 psi (2.23×10^4 MPa)

(ASTM C 469, modified)
Test conducted at a fluid consistency

Flexural Strength

3 days	1,000 psi (6.9 MPa)
7 days	1,050 psi (7.2 MPa)
28 days	1,150 psi (7.9 MPa)

(ASTM C 78)
Test conducted at a fluid consistency

Tensile Strength

3 days	490 psi (3.4 MPa)
7 days	500 psi (3.4 MPa)
28 days	500 psi (3.4 MPa)

(ASTM C 190)
Test conducted at a fluid consistency

Tensile Strength

Ultimate Tensile Strength and Bond Stress

Diameter	Depth	Tensile Strength	Bond Stress
5/8 in (15.9 mm)	4 in (101.6 mm)	23,500 in (10,575 mm)	2,991 in (20.3 mm)
3/4 in (19.1 mm)	5 in (127.0 mm)	30,900 in (13,905 mm)	2,623 in (18.1 mm)
1 in (25.4 mm)	6.75 in (171.5 mm)	65,500 in (29,475 mm)	3,090 in (21.3 mm)

(ASTM E 488, tests*)

*Average of 5 tests in ≥ 4,000 psi (27.6 MPa) concrete, using 125 ksi threaded rod in 2” (51 mm) diameter, damp, core-drilled holes.

Notes

1. Grout was mixed to a fluid consistency.
2. Recommended design stress: 2,275 psi (15.7 MPa).

3. For more detailed information regarding anchor bolt applications, contact Technical Service.
4. Tensile tests with headed fasteners were governed by concrete failure.

Shear Strength

Punching Shear Strength
3 by 3 by 11" (76 by 76 by 279 mm) beam

3 days	2,200 psi (15.2 MPa)	(Sika Method)
7 days	2,260 psi (15.6 MPa)	
28 days	2,650 psi (18.3 MPa)	

Shrinkage

Volume change

	% Change	% Requirement of ASTM C 1107	(ASTM C 1090)
1 day	> 0	0.0 – 0.30	
3 days	0.04	0.0 – 0.30	
14 days	0.05	0.0 – 0.30	
28 days	0.06	0.0 – 0.30	

Coefficient of Thermal Expansion

6.5 x 10^{-6} in/in/°F (11.7 x 10^{-6} cm/cm/°C) (ASTM C 531)
Test conducted at a fluid consistency

Freeze-Thaw Stability

Resistance to Rapid Freezing and Thawing
Durability Factor > 90% 300 Cycles (ASTM C 666, Procedure A)

Splitting tensile strength

3 days	575 psi (4.0 MPa)	(ASTM C 496) Test conducted at a fluid consistency
7 days	630 psi (4.3 MPa)	
28 days	675 psi (4.7 MPa)	

APPLICATION INFORMATION

Coverage

One 55 lb (25 kg) bag of SikaGrout®-928 grout mixed with 10.5 lbs (4.8 kg) or 1.26 gallons (4.8 L) of water (fluid consistency) provides approximately 0.50 ft3 (0.014 m3) of grout. Note: The water requirement may vary due to mixing efficiency, temperature, and other variables.

Set Time

	Plastic[1]	**Consistency Flowable**[2]	**Fluid**[3]	(ASTM C 191)
Initial set	2:30 hr:min	3:00 hr:min	4:30 hr:min	
Final set	4:00 hr:min	5:00 hr:min	6:00 hr:min	

1. 100–125% flow on flow table per ASTM C 230
2. 125–145% flow on flow table per ASTM C 230
3. 25 to 30 seconds through flow cone per ASTM C 939

Curing Conditions

Cure all exposed grout with an approved membrane curing compound compliant with ASTM C 309 or preferably ASTM C 1315. Apply curing compound immediately after the wet rags are removed to minimize potential moisture loss.

BASIS OF PRODUCT DATA

Results may differ based upon statistical variations depending upon mixing methods and equipment, temperature, application methods, test methods, actual site conditions and curing conditions.

ENVIRONMENTAL, HEALTH AND SAFETY

For further information and advice regarding transportation, handling, storage and disposal of chemical products, user should refer to the actual Safety Data Sheets containing physical, environmental, toxicological and other safety related data. User must read the current actual Safety Data Sheets before using any products. In case of an emergency, call CHEMTREC at 1-800-424-9300, International 703-527-3887.

APPLICATION INSTRUCTIONS

NOTES ON INSTALLATION

Forming
1. Forms should be liquid tight and nonabsorbent. Seal forms with putty, sealant, caulk or polyurethane foam. Use sufficient bracing to prevent the grout from leaking or moving.
2. Moderately sized equipment should utilize a head box to enhance the grout placement.
3. Side and end forms should be a minimum 1" (25 mm) distant horizontally from the equipment to be grouted to permit expulsion of air and any remaining saturation water as the grout is placed.
4. Leave a minimum of 2" between the bearing plate and the form to allow for ease of placement.
5. Eliminate large, non-supported grout areas wherever possible.
6. Extend forms a minimum of 1" (25 mm) higher than the bottom of the equipment being grouted.
7. Expansion joints may be necessary. Consult your local Sika field representative for suggestions and recommendations.

Temperature

1. The ambient and initial temperature of the grout should be in the range of 45 to 90 °F (7 to 32 °C) for both mixing and placing. For precision grouting, store and mix grout to produce the desired mixed-grout temperature. If bagged material is hot, use cold water, and if bagged material is cold, use warm water to achieve a mixed-product temperature as close to 70 °F (21 °C) as possible.
2. If temperature extremes are anticipated or special placement procedures are planned, contact your local Sika representative for assistance. 3. When grouting at minimum temperatures, see that the foundation, plate, and grout temperatures do not fall below 40 °F (7 °C) until after final set. Protect the grout from freezing (32 °F or 0 °C) until it has attained a compressive strength of 3,000 psi (21 MPa) in accordance with ASTM C 109.

Recommended Temperature Guidelines for Precision Grouting

	Minimum °F (°C)	Preferred °F (°C)	Maximum °F (°C)
Foundation and plates	45 (7)	50–80 (10–27)	90 (32)
Mixing water	45 (7)	50–80 (10–27)	90 (32)
Grout at mixed and placed temp.	45 (7)	50–90 (10–32)	90 (32)

SURFACE PREPARATION

1. Steel surfaces must be free of dirt, oil, grease, or other contaminants.
2. The surface to be grouted must be clean, SSD, strong, and roughened to a CSP of 5–9 following ICRI Guideline 310.2 to permit proper bond.
3. When dynamic, shear or tensile forces are anticipated, concrete surfaces should be chipped with a “chisel-point” hammer, to a roughness of (plus or minus) 3 ⁄8" (10 mm). Verify the absence of bruising following ICRI Guideline 210.3.
4. Concrete surfaces should be saturated (ponded) with clean water for 24 hours just before grouting.
5. All freestanding water must be removed from the foundation and bolt holes immediately before grouting.
6. Anchor bolt holes must be grouted and sufficiently set before the major portion of the grout is placed.
7. Shade the foundation from sunlight 24 hours before and 24 hours after grouting.

MIXING

By using the minimum amount of water to provide the desired workability, maximum strength will be achieved. Whenever possible, mix the grout with a mortar mixer or an electric drill with a paddle such as ICRI 320.5 type A, D, E, F, G or H. Put the measured amount of potable water into the mixer, add grout, then mix till a uniform consistency is attained. Do not use water in an amount or a temperature that will cause bleeding or segregation. Note: The water requirement may vary due to mixing efficiency, temperature, and other variables.

1.Place estimated water (use potable water only) into the mixer, then slowly add the grout. For a fluid consistency, start with 9 lbs (4 kg) (1.1 gallon [4.2L]) per 55 lb bag.
2.The water demand will depend on mixing efficiency, material, and ambient-temperature conditions. Adjust the water to achieve the desired flow. Recommended flow is 25–30 seconds using the ASTM C 939 Flow-Cone Method. Use the minimum amount of water required to achieve the necessary placement consistency.
3.Moderately sized batches of grout are best mixed in one or more clean mortar mixers. For large batches, use ready-mix trucks and 3,300 lb (1,500 kg) bags for maximum efficiency and economy.
4.Mix grout between 3 and 5 minutes after all material and water is in the mixer until a homogenous consistency is achieved. Use mechanical mixer only.
5.Do not mix more grout than can be placed in approximately 30 minutes.
6.Transport by wheelbarrow or buckets or pump to the equipment being grouted. Minimize the transporting distance.
7.Do not retemper grout by adding water and remixing after it stiffens.
8.Do not add plasticizers, accelerators, retarders, or other additives.
9.For placements greater than 6" (152 mm) in depth, product should be extended with aggregate. Aggregate extension is dependent upon the grout type, placement, application requirements, and is typically required for placement depths beyond the limitation of the neat

material. The aggregate should be washed, graded, saturated, surface-dry (SSD), high-density, free from deleterious materials, and comply with the requirements of ASTM C 33. Consult Sika Technical Service for additional guidance.

APPLICATION

Placement

1. Always place grout from only one side of the equipment to prevent air or water entrapment beneath the equipment. Place SikaGrout®-928 in a continuous pour. Discard grout that becomes unworkable. Make sure that the material fills the entire space being grouted and that it remains in contact with plate throughout the grouting process.
2. Immediately after placement, trim the surfaces with a trowel and cover the exposed grout with clean wet rags (not burlap). Keep rags moist until grout surface is ready for finishing or until final set.
3.The grout should offer stiff resistance to penetration with a pointed mason's trowel before the grout forms are removed or excessive grout is cut back. After removing the damp rags, immediately coat with a recommended curing compound compliant with ASTM C 309 or preferably ASTM C 1315.
4.Do not vibrate grout. Use steel straps inserted under the plate to help move the grout. 5.Minimum placement thickness is 1" (25 mm). Consult your Sika representative before placing lifts more than 6" (152 mm) in depth.

CLEANING

Waste Disposal Method

This product when discarded or disposed of, is not listed as a hazardous waste in federal regulations. Dispose of in a landfill in accordance with local regulations. For additional information on personal protective equipment, first aid, and emergency procedures, refer to the product Safety Data Sheet (SDS) on the job site or contact the company at the address or phone numbers given below

OTHER RESTRICTIONS

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